

Fractions, Decimals & Percentages

PAPER C — HARD

Name _____ Date _____ Score _____

Before you start

- No calculator. Some of these have a short route and a long route — the short route is usually a single multiplier.
- If the amount you are given is described as being **after** a change (“after a discount”, “including tax”, “20% more than”), you are dividing, not multiplying.
- Successive percentage changes **multiply**. They never add. At least one question on this paper is built on that.
- Percentage change is always measured against the **original** value, not the new one.
- Check every reverse percentage by working forwards again. It takes ten seconds and catches almost every slip.

SECTION A — TECHNIQUE

- 1 Write $5/12$ as a decimal, using recurring notation where appropriate. [2]

Answer

- 2 Work out $2\frac{1}{3} - 1\frac{3}{4}$, giving your answer in its simplest form. [2]

Answer

- 3 After a 20% discount, a coat costs ₹1960. Calculate the price before the discount. [2]

Answer ₹

- 4 A price is increased by 25% and the new price is then decreased by 20%. Show by calculation that the final price is the same as the original price. [3]

SECTION B — PROBLEMS

- 5 $\frac{3}{5}$ of a number is 42. [3]

(a) Find the number. [2]

(b) Hence find 25% of that number. [1]

- 6 A jacket has a marked price of ₹3600. Two shops sell it. [4]

Shop P: 40% off the marked price.

Shop Q: $\frac{1}{3}$ off the marked price, then a further 10% off that reduced price.

(a) Find the price at each shop. [2]

(b) State which shop is cheaper, and by how much. [1]

(c) Find the single percentage discount that Shop Q is actually offering. [1]

- 7 A student writes: “If I increase 80 by 10% and then decrease the result by 10%, I get back to 80.” [3]

(a) Explain why the student is wrong. [1]

(b) Calculate the correct final value. [2]

- 8 Write these four numbers in order, smallest first. You must show the values you compared, and state clearly if any two of them are equal. [3]

0.66 2/3 65.5% 33/50

Answer

- 9 A shopkeeper buys a fan for ₹450 and sells it for ₹549. [4]

(a) Calculate the percentage profit. [2]

(b) What selling price would give him a profit of exactly 30%? [2]

10 In a school, $\frac{5}{8}$ of the students are girls. $\frac{2}{5}$ of the girls play a sport. [4]

(a) What fraction of the whole school are girls who play a sport? [2]

(b) The school has 480 students. How many girls play a sport? [1]

(c) What percentage of the school is this? [1]
